C# Programming – Comprehensive Syllabus

Course Overview

This course provides a comprehensive introduction to C# programming using the .NET platform. Students will learn programming fundamentals, object-oriented programming (OOP), exception handling, file handling, collections, LINQ, database connectivity, Windows application development, and modern application development practices.

Duration: 8–12 Weeks (50–70 Hours)

Prerequisites:

- Basic computer knowledge
- No prior programming experience required

Module 1: Introduction to C# & .NET (Week 1)

Topics

- Introduction to C#
- .NET Framework & .NET Platform
- Installing Visual Studio
- Structure of a C# Program
- Variables and Data Types
- Constants
- Input and Output
- Operators
- Type Conversion
- Comments and Naming Conventions

Practical Exercises

- Hello World Program
- Basic Calculator
- Unit Conversion Programs

Module 2: Decision Making & Looping (Week 2)

Topics

- if, if-else, Nested if
- switch Statement
- for Loop
- foreach Loop
- while Loop
- do-while Loop
- break and continue

Practical Exercises

- Grade Calculator
- Leap Year Checker
- Multiplication Table
- Number Guessing Game

Module 3: Methods & Arrays (Week 3)

Topics

- Methods
- Parameters and Return Types
- Method Overloading
- Optional Parameters
- Arrays
- Multidimensional Arrays
- Strings
- String Manipulation

Practical Exercises

- Matrix Operations
- String Processing
- Recursive Programs
- Menu-Driven Applications

Module 4: Object-Oriented Programming (Week 4)

Topics

- Classes and Objects
- Constructors
- Destructors
- Properties
- Encapsulation
- Access Modifiers
- Static Members
- this Keyword

Practical Exercises

- Student Management System
- Employee Management
- Bank Account Simulation

Module 5: Advanced OOP Concepts (Week 5)

Topics

- Inheritance
- Polymorphism
- Method Overriding
- Abstract Classes
- Interfaces
- Sealed Classes
- Partial Classes

Practical Exercises

- Vehicle Management System
- Library Management System
- Shape Calculator

Module 6: Exception Handling, Collections & LINQ (Week 6)

Topics

- Exception Handling (try, catch, finally, throw)
- Generic Collections
- List
- Dictionary
- Queue
- Stack
- LINQ Basics
- Lambda Expressions

Practical Exercises

- Exception Handling Programs
- Collection-Based Applications
- LINQ Queries

Module 7: File Handling & Database Connectivity (Week 7)

Topics

- File I/O Operations
- StreamReader & StreamWriter
- Binary Files
- SQL Server Basics
- ADO.NET
- CRUD Operations
- Database Connectivity

Practical Exercises

- Student Database
- Employee Records
- CRUD Application

Module 8: Windows Forms & Introduction to ASP.NET (Week 8)

Topics

- Windows Forms Basics
- Event Handling
- GUI Controls
- Menu Systems
- Introduction to ASP.NET
- MVC Architecture (Overview)
- Web Application Basics

Practical Exercises

- Calculator Application
- Student Registration Form
- Basic Web Application

Module 9: Mini Project (Week 9)

Choose one:

- Library Management System
- Banking Management System
- Student Information System
- Hospital Management System
- Inventory Management System
- Employee Payroll System
- Hotel Reservation System

Module 10–12: Capstone Project

Complete Application Development Project

- Requirement Analysis

- Object-Oriented Design
- Database Design
- User Interface Development
- CRUD Implementation
- Exception Handling
- Testing and Debugging
- Documentation
- Final Demonstration

Tools & Software

- Microsoft Visual Studio
- .NET SDK
- SQL Server Express
- SQL Server Management Studio (SSMS)
- Git & GitHub

Core Concepts Covered

- C# Programming Fundamentals
- Object-Oriented Programming (OOP)
- Methods and Arrays
- Exception Handling
- Generic Collections
- LINQ
- File Handling
- Database Connectivity (ADO.NET)
- Windows Forms
- Introduction to ASP.NET

Assessment Scheme

Component	Weightage
Assignments	20%
Lab Exercises	20%

Quizzes	10%
Mini Project	20%
Capstone Project	30%

Learning Outcomes

By the end of the course, learners will be able to:

- Understand the fundamentals of C# programming and the .NET ecosystem.
- Develop console and Windows-based applications using C#.
- Apply object-oriented programming principles including encapsulation, inheritance, polymorphism, and abstraction.
- Use collections and LINQ for efficient data processing.
- Handle exceptions and perform file operations for persistent data storage.
- Connect applications to SQL Server databases using ADO.NET and implement CRUD operations.
- Build user-friendly desktop applications and understand the basics of ASP.NET web development.
- Debug, test, and optimize C# applications using industry-standard tools.
- Develop real-world software applications following modern programming and software development practices.